

Project Title: Customer Segmentation for Brazilian Online Retail Using Machine Learning

1. Problem Statement

The contemporary retail industry encounters a wide variety of customer behaviours and purchasing preferences that render traditional, uniform marketing campaigns inefficient. To optimize resource allocation and improve customer satisfaction and retention, online retailers must leverage advanced analytics to identify distinct customer segments. This project aims to develop a data-driven customer segmentation model based on the Brazilian E-Commerce Public Dataset, facilitating precise targeting and personalized marketing strategies that boost business performance.

2. Business Goal

The primary goal is to develop a customer segmentation model using machine learning that accurately identifies distinct and actionable customer groups based on their purchasing behaviour, payment patterns, and review history. This segmentation model will enable the marketing team to deploy personalized marketing campaigns and tailored retention strategies aimed at increasing customer engagement and improving repeat purchase rates.

3. Data Source

This project utilizes the Brazilian E-Commerce Public Dataset provided by Olist, accessible via Kaggle. The dataset comprises approximately 100,000 orders along with associated customer profiles, payment details, product information, seller data, and customer reviews. This comprehensive dataset offers a realistic and multifaceted view of customer interactions in a large marketplace setting.

- Source: Kaggle
- URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

4. Tools & Technologies

- **Programming Language:** Python
- **Key Libraries:** pandas, numpy, scikit-learn, matplotlib, seaborn
- **Development Environment:** Google Colab, Jupyter Notebook, VS Code
- **Business Intelligence Tools:** Tableau or Power BI for interactive dashboards and reports

5. Project Workflow

The project will follow a systematic data science lifecycle consisting of:

Phase	Activities & Purpose
Data Acquisition & Integration	• Download dataset using Kaggle API
	• Merge tables (orders, customers, payments, reviews)
	• Validate schema and join keys
Data Cleaning & Preprocessing	• Handle missing values, duplicates, and outliers
	• Normalize formats (e.g., dates, categories)
	• Encode categorical variables
Exploratory Data Analysis (EDA)	• Visualize distributions and trends
	• Identify behavioural patterns and anomalies
	• Use matplotlib/seaborn for insights
Feature Engineering	• Derive RFM metrics (Recency, Frequency, Monetary)
	• Aggregate review scores and payment types
	• Create customer-level features
Clustering Model Development	• Apply K-Means, Hierarchical Clustering, DBSCAN
	• Tune parameters using Elbow Method, Silhouette Score
	• Compare cluster stability
Cluster Evaluation & Interpretation	• Validate clusters using internal metrics
	• Map clusters to business personas (e.g., loyal, high-value, dormant)
	• Document actionable insights
Dashboard & Reporting	• Build interactive dashboards in Tableau/Power BI
	• Summarize findings for marketing and strategy teams
	• Include cluster profiles and campaign suggestions

6. Data Extraction

Automated data extraction will be performed using Kaggle's API with scripts encapsulated in the Python notebook file. This notebook will streamline the downloading, preprocessing, and merging of dataset components into a clean, analysis-ready structure, ensuring reproducibility and ease of collaboration.

7. Schema/Data Dictionary

The project will include an Excel document titled Schema.xlsx that thoroughly describes all dataset features, including field names, data types, descriptions. This document will serve as a reference to ensure consistent understanding and usage of data attributes throughout the project.

